Hysteresis curves of ferro- and diamagnetic materials

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The hysteresis curves of two $FeNi$-alloys (PERMENORM 5000 H2 and PERMENORM 5000 Z), wood and flatsteel (ST 37 K) were measured. These were used to determine the remanence as well as the coercivity for the metalic materials and the permeability of wood. Additionally the saturation flux and the hysteresis losses of both the $FeNi$-alloys and iron are calculated. The measured values for remanence and coercivity are $B_{R,ST\,37\,K} = (548.74 \pm 0.93)mT$, $B_{R,5000\,H2} = (653.9 \pm 2.3)mT$, $B_{R,5000\,Z} = (1427.8 \pm 2.4)mT$, $H_{C,ST\,37\,K} = (358 \pm 17)\frac{A}{m}$, $H_{C,5000\,H2} = (3.476 \pm 0.024)\frac{A}{m}$, $H_{C,5000\,Z} = (13.879 \pm 0.012)\frac{A}{m}$ and for the saturation flux density $B_{S,5000\,H2} = (1154.144 \pm 0.092)mT$, $B_{S,5000\,Z} = (1448.87 \pm 0.14)mT$. The relative permeability of wood was measured to be $\mu_{R,Holz} = 1.816507 \pm 0.000050$, which points to a problem with the experimental setup. Furthermore the irreversibility of the hysteresis curve and the process of demagnetization are demonstrated on flatsteel.

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1 Introduction

According to the Bohr–van Leeuwen theorem, condensed matter should not exhibit any magnetization under classical assumptions and thus requires a theory of firmly quantum mechanical nature. Yet, despite being a purely quantum mechanical effect, magnetism exhibits a large variety of well known and studied macroscopic behaviours - from keeping notes stuck to ones fridge or pinpointing the direction of earths magnetic poles to heavy duty lifting with electro-magnetic cranes. An especially important process is magnetization itself, since in our modern digitized world data management relies heavily on the duration and facility of magnetization in the used materials, e.g., in hard drives. In the following, some of those macroscopic properties shall be studied through coils with different cores.

2 A simple model for macroscopic magnetic phenomena

In order to have no reliance on quantum mechanics one can consider a simple model, which still manages to make correct experimental predictions[1]. It states that a magnetic material is made up of a multitude of small magnetic dipoles, the so called elementary magnets, which on a microscopic level tend to form small areas of equal alignment called magnetic domains. The overall magnetization of the material thus depends on the average direction of all the elementary magnets, e.g. a piece of iron where the elementary magnets are unaligned would be demagnetized.
If one now applies a strong enough external magnetic field, all elementary magnets align in its direction - this state is called saturation magnetization, since the material cannot be magnetized any further. Turning the field off again will not demagnetize the material, since most of the elementary magnets stay aligned due to a preferable internal energy. This effect is called remanence and its associated with a specific flux density B_R - this flux density is usually refered to as remanence. Applying a magnetic field in the opposite direction and increasing its strength one will reach a certain field strength H_C, where all elementary magnets are unaligned and the overall magnetization becomes zero. H_C is called the coercivity and its main importance is that it signifies the fact, that it takes work to realign the elementary magnets. All of the named values are very important material characteristics, which determine their magnetic behaviour and thus their possible applications in industry and science. The measurements of H_C and B_R were made under assumption of the elementary magnet model.

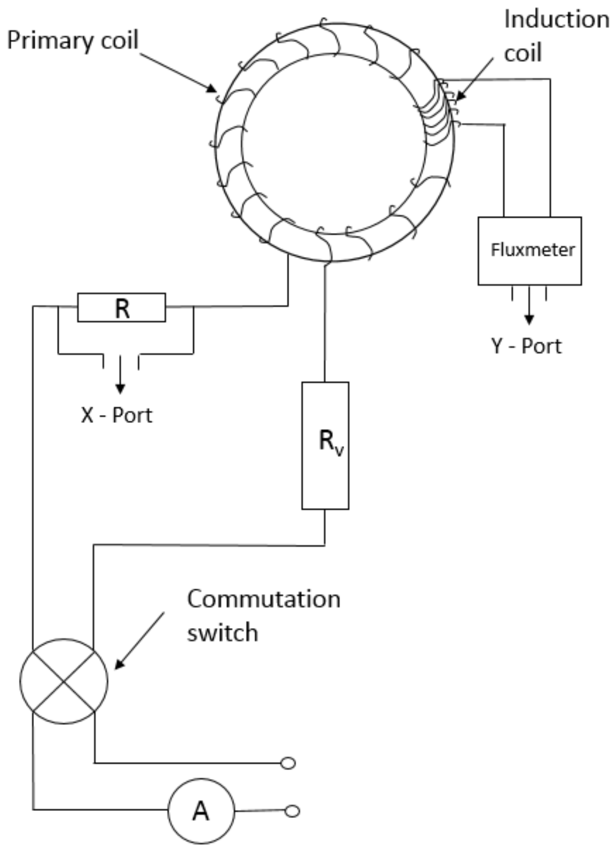


Figure 1: Experimental setup to measure magnetic properties of different coil materials. The measured core material is the primary coil. The two voltages were recorded and the magnetic flux density and magnetic field strength were calculated from them[2]

3 Hysteresis curves of $FeNi$-alloys and wood

An experimental setup according to (Figure 1) was built. It features an adjustable maximum current, the voltages U_R and U_F were measured and integrated via a fluxmeter. Measuring the desired hysteresis curves was achieved by starting out at a certain maximum current, which is then decreased to zero together with the applied voltage U_R. Then the polarity is switched and I is increased to its maximal value again- this process is then repeated one more time. The curve is plotted via a computer program, which calculates the magnetic flux $B(H)$ via the equations

$$H = \frac{2N_1U_R}{\pi(d_a + d_i)R} \tag{1}$$

$$B = \frac{2U_F m_F}{\xi h N_2(d_a - d_i)} \tag{2}$$

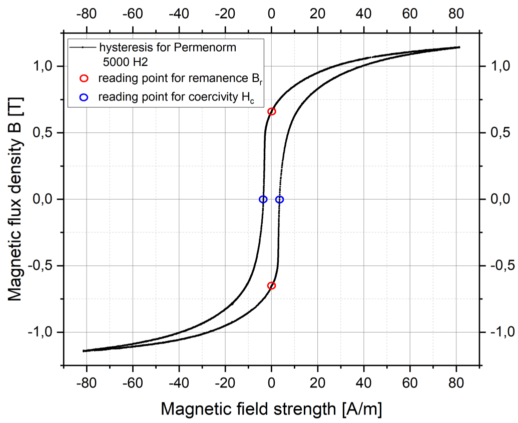


Figure 2: Hysteresis curve for $FeNi$-alloy PERMENORM 5000 H2 with reading points at the axis intersections to determine the remanence and coercivity.

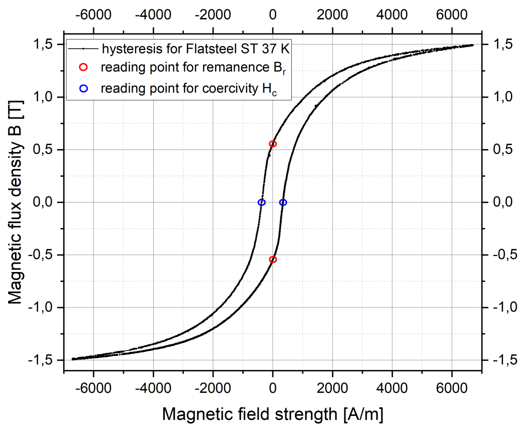


Figure 3: Hysteresis curve for flatsteel ST 37 K with reading points at the axis intersections to determine the remanence and coercivity.

The specific parameters for the given coils can be found in Table 1. Flux density B and field strength H were calculated with equations (1) and (2) for the two $FeNi$-alloys, flatsteel and wood.
Figure 7 shows a linear progression. This hints at a constant magnetic permeability which is given by

$$\mu = \frac{B}{H}. \tag{3}$$

Thus μ describes the magnetization of a material in an external field. The permeability is also often expressed through the relative permeability μ_R which is just the normal permeability divided by the permeability of free space

$$\mu_R = \frac{B}{\mu_0 H}. \tag{4}$$

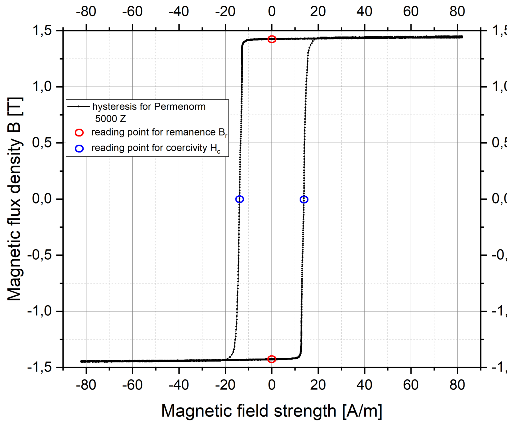


Figure 4: Hysteresis curve for $FeNi$-alloy PERMENORM 5000 Z with reading points at the axis intersections to determine the remanence and coercivity.

The magnetic behaviour is thus often classified by whether or not μ_R is greater or smaller than 1, with ferro- and paramagnetic materials posessing a $\mu_R > 1$ and diamagnetic materials with a $\mu_R < 1$.
The ratio $\frac{B}{H}$ is given by the slope of Figure 7, thus $\mu_R = 1.816507 \pm 0.000050$ [3]. Since the magnetic properties of wood mostly arise from the diamagnetic water within, one would expect a relative permeability $\mu_R < 1$. The obvious discrepancy could be explained by the other three magnetized cores nearby, which disrupt the measurement with their own magnetic fields. This interference could have been prevented or at least lessened by either demagnetizing the cores, or dismantling the pre-assembled coil setup.

	5000 H2	5000 Z	Flatsteel	Wood
d_a	$260mm$	$260mm$	$260mm$	$260mm$
d_i	$220mm$	$220mm$	$220mm$	$220mm$
N_1	605	610	1010	1010
N_2	80	56	53	250
h	$30mm$	$30mm$	$30mm$	$30mm$
ξ	90%	90%	100%	100%
R	$66.6\,\Omega$	$66.6\,\Omega$	$0.2\,\Omega$	$0.2\,\Omega$

Table 1: Coil specific parameters[4]

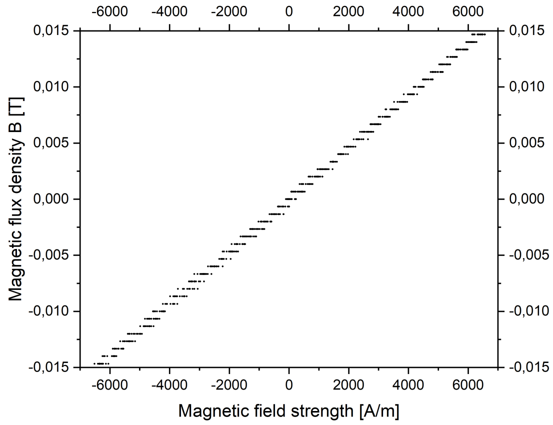


Figure 5: Hysteresis curve for the wooden core at a less sensitive measuring range setting for the fluxmeter.

As one can see in Figure 5, due to the less sensitive measuring range setting of the fluxmeter at $m_f = 100\,ms$, the voltage resolution is too low. This results in a step-like slope. Decreasing the measuring gauge to $m_f = 5\,ms$ nullifies this effect and one can find a more comprehensive result, as shown in Figure 6. But one can clearly detect a time dependent drift of the magnetic flux density. This is caused by the capacity in the fluxmeter which discharges during the measurement process. The voltage decay of a capacity is an exponential process, but in this case a linear approximation is sufficient. The linear time dependent voltage decay is computed by the quotient of the flux density and the time between the two data points at the same magnetic flied strength. In the corrected Figure 7 a linear fit is used to compute the relative permeability of wood, which is $\mu_R = 1.816507 \pm 0.000050$.

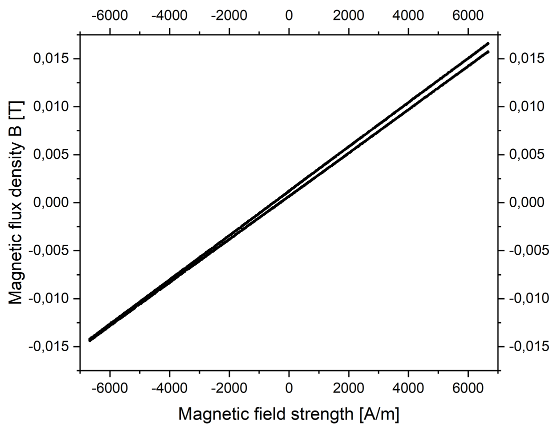


Figure 6: Hysteresis curve for the wooden core with a more appropriate measuring range setting for the fluxmeter.

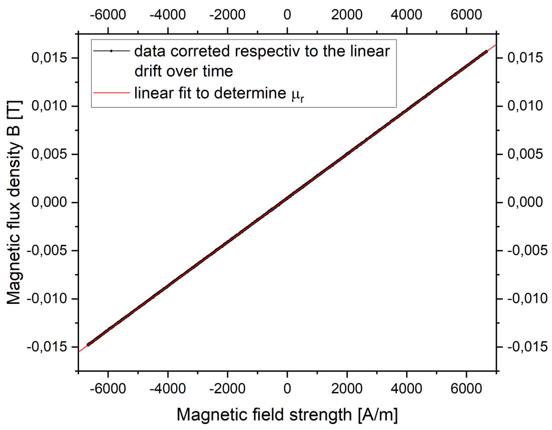


Figure 7: Hysteresis curve for the wooden core with a more appropriate measuring range setting for the fluxmeter and corrected data. To compute the relative permeability μ_R a linear fit is plotted.

4 Remanence and coercivity

Both coercivity and remanence are determined through a graphical evaluation of the hysteresis curves, which rely on finding the curves point of central symmetry and taking it as a coordinate origin. The intersections of the curve with the coordinate axis are B_R and H_C respectively. Table 2 shows the values gained from Figure 2 to Figure 4 - the shown values are the mean of the two corresponding sections. The errors are given by the deviation from the mean and the intersection.

Material	B_R in mT	H_C in $\frac{A}{m}$
Flatsteel	$548.74 \pm 0,93$	358 ± 17
5000 H2	653.9 ± 2.3	3.476 ± 0.024
5000 Z	1427.8 ± 2.4	13.879 ± 0.012

Table 2: Values of remanence and coercivity for the corresponding materials

As one can easily see in Table 2, flatsteel exhibits the highest coercivity of the examined materials. This is expected since usually iron posseses a high coercivity of 6 to 12 $\frac{A}{m}$ [5], which in the case of flatsteel should be lowered by the added carbon. It also features the lowest remanence of the three metals, meaning its potential uses are applications where a not too strong permanent magnet with high resistance to external fields is required. The permalloys, which are known to be soft magnetic materials, exhibit a low coercivity. The industrial data sheet supplied by the producer of the permalloys [6] gives for the 5000 H2 core a coercivity of 5 $\frac{A}{m}$ - for 5000 Z no values were given, since this core model might not be in production any more. In an older book [7], the coercivity of 5000 Z is given as $H_C = 9.6\,\frac{A}{m}$ The values for flatsteel and 5000 H2/Z are thus not in agreement with the given reference values with the latter two being at least of the same order of magnitude. The measured coercivity for flatsteel, in contrast, is off by two orders of magnitude. As for flatsteel one should also keep in mind, that the cited reference value for iron does not refer to a specific industrial application such as the used ring core, but is instead a more general value which refers to the raw material.

5 Saturation flux

To measure the saturation flux for PERMENORM 5000 H2, the maximum current has to be increased to $I = 2.3\,A$, since (Figure 2) shows no parallelity to the x-axis and saturation is not yet reached. The new measurement results can be seen in Figure 8. Since the saturation flux for PERMENORM 5000 H2 is not reached, the value for the saturation flux itself is not significant. To compute the saturation flux B_s for each material, the mean value and the corresponding standard deviation of 100 different data points at each end of the hysteresis are calculated. The results are shown in Table 3.

Material	B_S in mT
5000 H2	1154.144 ± 0.092
5000 Z	1448.87 ± 0.14

Table 3: Values of the saturation flux for both PERMENORM $FeNi$-alloys

H2 shows a distinctively higher saturation density than Z. So a higher magnetic field is necessary to magnetize the PERMENORM 5000 H2 completely. As seen in Figure 8, it was not possible to reach saturation, which would be indicated by a horizontal progression. The measured value for B_S for PERMENORM 5000 H2 is thus expected to be smaller than the reference values.

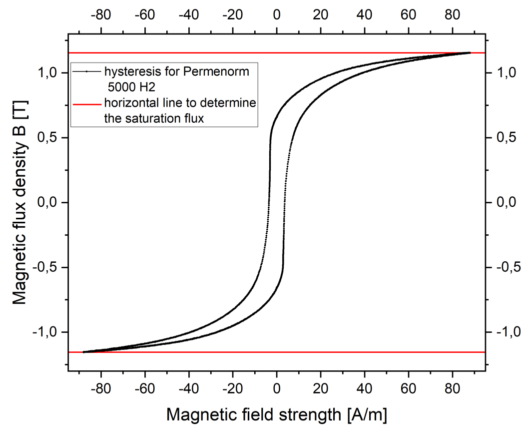


Figure 8: Hysteresis curve for $FeNi$-alloy PERMENORM 5000 H2 with two horizontal red lines to indicate the saturation flux.

In Figure 9 the saturation flux is reached due to the horizontal progression of the hysteresis curve. In the product data sheets for PERMENORM 5000 H2, the saturation flux is given as $B_S = 1.55\,T$ [6]. Generally, Fe has a saturation flux of $B_{S,Fe} = 2,15\,T$ and $FeNi$-alloys with around 50% Ni have a saturation flux B_S of $1.3\,T$ to $1.5\,T$ depending on the composition of the material [5]. So the measured value for 5000 Z is in the expacted range for $FeNi$-alloys. The value for 5000 H2 is, as expected, lower than the reference value.

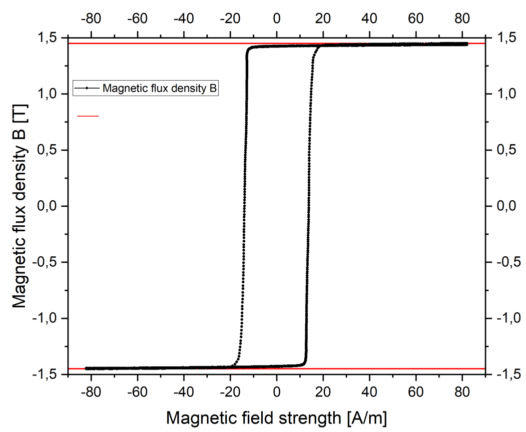
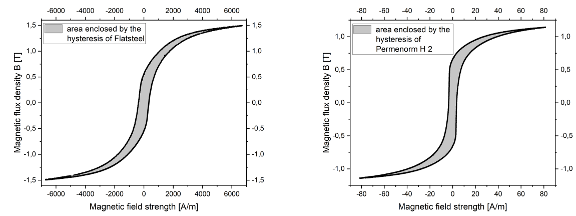


Figure 9: Hysteresis curve for $FeNi$-alloy PERMENORM 5000 Z with two horizontal red lines to indicate the saturation flux.

6 Hysteresis loses



(a) **(b)**

Figure 10: Numerical computed area in the hysteresis curve of
(a) Flatsteel ST 37 K
(b) $FeNi$-alloy PERMENORM 5000Z

The area A enclosed in the hysteresis curve is proportional to the energy loss through thermal heat dissipation during remagnetization. The energy-loss density (eld) is thus given by the area A and the density ρ

$$\omega = \frac{A}{\rho}. \tag{5}$$

The area in the hysteresis curve of flatsteel and $FeNi$-alloy 5000 H2 was determine numerically (Figure 10). The datapoints themselves have no errors, thus the calculated area A has no errors. The reference values for the densities given in [1] assumed to be free of errors. Thus ω also has no error - no arbitrary error estimations were made.

Material	eld in $\frac{mJ}{kg}$	density ρ in $\frac{kg}{m^3}$
5000 H2	2.640	8250
ST 37 K	269.3	7800

Table 4: Values of the energy loss per kg because of remagnetization for the $FeNi$-alloy PERMENORM 5000 H2 and flatsteel ST 37 K computed by the density ρ and the area enclosed by the hysteresis curve.

The results for the $FeNi$-alloys and flatsteel are shown in Table 4. As one can easily see the iron core has higher energy losses during remagnetization than the iron-nickel-alloys. Thus it would be, as mentioned in the introduction, not suited for applications where frequent remagnetization is required.

7 Irreversibility of hysteresis curves

To demonstrate the irreversibility, a normal hysteresis curve is measured for a iron core with an addition. When zero current is reached, one increases the current to $0.5\,A$ and back again to $0\,A$ before each use of the polarity switch. This was done for the upper and the lower branch. The result is shown in Figure 11. The figure indicates a change in the progression of the hysteresis curve.

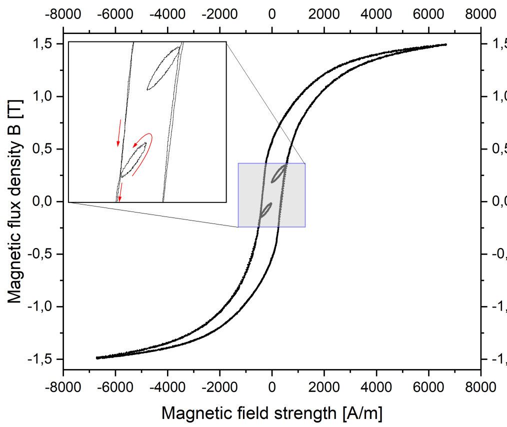


Figure 11: Hysteresis curve for Flatsteel ST 37. To achieve such a curve, a normal hysteresis curve is measured. On a second circulation the current was limited to $0,5\,A$ and than brought back to $0\,A$. To underline the irreversibility, the resulting change of the curve was enhanced and the circulation sense is indicated by red arrows.

Since the small loops diverge from the original characteristical hysteresis curves the magnet exhibits a typical property of irreversible systems, namely being unable to reach its starting point without tracing back its original path - in this case reaching the same magnetic flux density by increasing or decreasing the magnetic field strength. Magnetic irreversibility itself is a complex phenomenon resulting from a multitude of causes such as the movement of magnetic domain walls or the potential anisotropy of a given material. Explained in terms of the simplified model, the inertia of the elementary magnets would be the main cause of irreversibility.

8 Demagnetization of a iron core

The iron core is demagnetized through manipulation of the maximum current. Essentially one measures normal hysteresis curves, with a decreasing maximum current after each cycle, the characteristical inner hysteresis curve is created. Once the maximum current is zero the core is demagnetized as shown in Figure 12. This procedure is able to demagnetize the core due to the already discussed irreversibility of hysteresis curves. With each cycle - and decreased maximum current - less and less elementary magnets align with the magnetic field until at zero ampere their orientations cancel out, resulting in a almost vanishing macroscopic magnetization. Nevertheless in the experimental execution was not possible to reach exactly 0 magnetic flux. A small amount of magnetic flux of $88\,mT$ remains. A better result could be achieved by decreasing the current even slower and continual, not after every full cycle. Besides that, the demagnetized coil is surrounded by other magnetized coils, interfering in the demagnetization process.

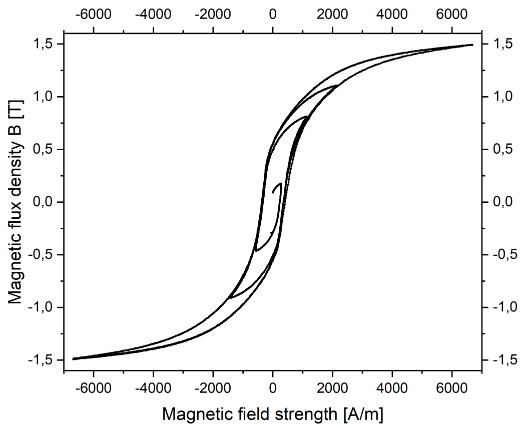


Figure 12: Inner hysteresis curve for flatsteel ST 37 created by decreasing the maximung current after each cycle.

9 Conclusion

By measuring hysteresis curves for flatsteel, two different $FeNi$-alloys and wood, it was possible to determine the coercivity and remanence of the metals, as well as the permeability of the latter. The corresponding values are listed in Table 5. These values are roughly representative of the magnetic properties of the given materials. This is not the case for the relative permeability of wood, since its $\mu_R = 1.816507 \pm 0.000050$, which is way to high for a diamagnetic material. The results overall show, that the measurement is quite susceptible to interference from outside sources and the other components of the experiment itself, especially for the flatsteel core, which deviates by two orders of magnitude from the reference values [5]. As such it is advisable to put a greater distance between the used coil cores or to always demagnetize them beforehand. Another important material specific value is the saturation flux, which is the highest possible magnetic flux inside a given material (see Table 5). The measurements did not agree with the reference values in their error margin. Especially for PERMENORM 5000 H2 since saturation wasn't reached, thus being much lower than the given reference value. Since during magnetization energy is lost through heat dissipation, the energy-loss-density was determined with practical applications in mind for 5000 H2 and flatsteel (ST 37 K). The values for the corresponding metals are $0.00264\frac{J}{kg}$ and $0.269\frac{J}{kg}$. As such the a iron core is not suited for use in setups where frequent remagnetization is required. Additionally the irreversibility of hysteresis curves was demonstrated and applied in the demagnetization of an iron core.

	5000 H2	5000 Z
H_C in $[\frac{A}{C}]$	3.476 ± 0.024	13.476 ± 0.024
B_R in $[mT]$	653.9 ± 2.3	1427.8 ± 2.4
B_s in $[mT]$	1154.144 ± 0.092	1448.87 ± 0.14
	ST 37 K	
H_C in $[\frac{A}{C}]$	358 ± 17	
B_R in $[mT]$	548.74 ± 0.93	

Table 5: Measured values for the coercivity H_C, remanence B_R and saturation flux B_S for the materials PERMENORM 5000 H2, PERMENORM 5000 Z and the flatsteel ST 37 K.

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